
بسم الله الرحمن الرحيم







REMOTE SENSING - AN OVERVIEW







“Man must rise above the Earth

To the top of the atmosphere and beyond

For only thus will he fully understand the world in which he lives.”

Socrates

Maps

- Maps are critical elements for national development. When the East India Company obtained control of India, one of the first orders of business was to survey the country
- Much of this need was driven by fiscal needs, although these topographic maps were used for other purposes

Remote Sensing

- **Remote Sensing ?**

Remote Sensing

- **Remote:** Not in contact with, Away from
- **Sensing:** The act of observation



Remote Sensing

- It is the acquisition of information about a phenomenon while still at some distance from it (not in contact)
- The technique of obtaining information about objects through the analysis of data collected by special instruments that are not in physical contact with the objects of investigation, and can be regarded as reconnaissance from a distance
- The act of seeing is, in effect," remote sensing," and the eye is a "remote sensor"

Remote Sensing

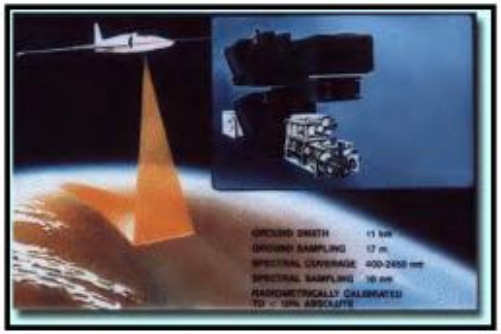
- **Mapping**: collection of thematic and quantitative baseline data
- **Measuring**: more rigorous mapping process by quantifying and documenting attributes of phenomena
- **Modeling**: a process of describing a system under study through precise and typically mathematical relations of inputs and outputs and to simulate its present, past or future behavior
- **Monitoring**: regular assessment of the conditions by recording the changes in natural phenomena and human activities
- These functions require valid, up-to date, spatially and temporally explicit natural resource data for decision-making

Milestone in the History of Remote Sensing

1840- 1860s | 1890s | 1900s | 1960

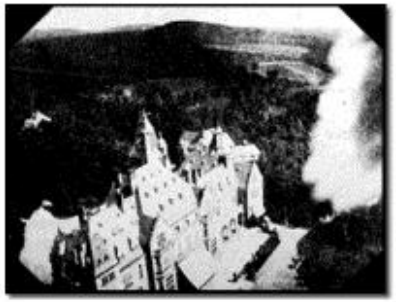
















Balloons → Pigeons → Airplanes → Satellites

RS of the Earth

- 1957 – Sputnik—Soviet Union-start of space age, turned the thinking to consideration of how the satellite might be useful
- 1958 – NASA was formed, Could aerospace technology be used to better manage the Earth's natural and man made resources?
- 1960 – TIROS-1 (Television Infrared Observation Satellites),basically a weather sat, principally to provide image of clouds cover over wide area, first use of term *"remote sensing"*

RS of the Earth

- 1960 - 1980 – some 40 earth observational satellites launched (Landsat, SPOT)
- Late 1990s, high resolution commercial satellites (IKONOS)
- 2000-onwards, new era of very high resolution sat- QuickBird, orb view, GeoEye,WorldView
- Any one you know about???
- Google Earth 2005: a virtual representation of the Earth's surface as a composite of varied digital images

Data and Information ?

Data and Information

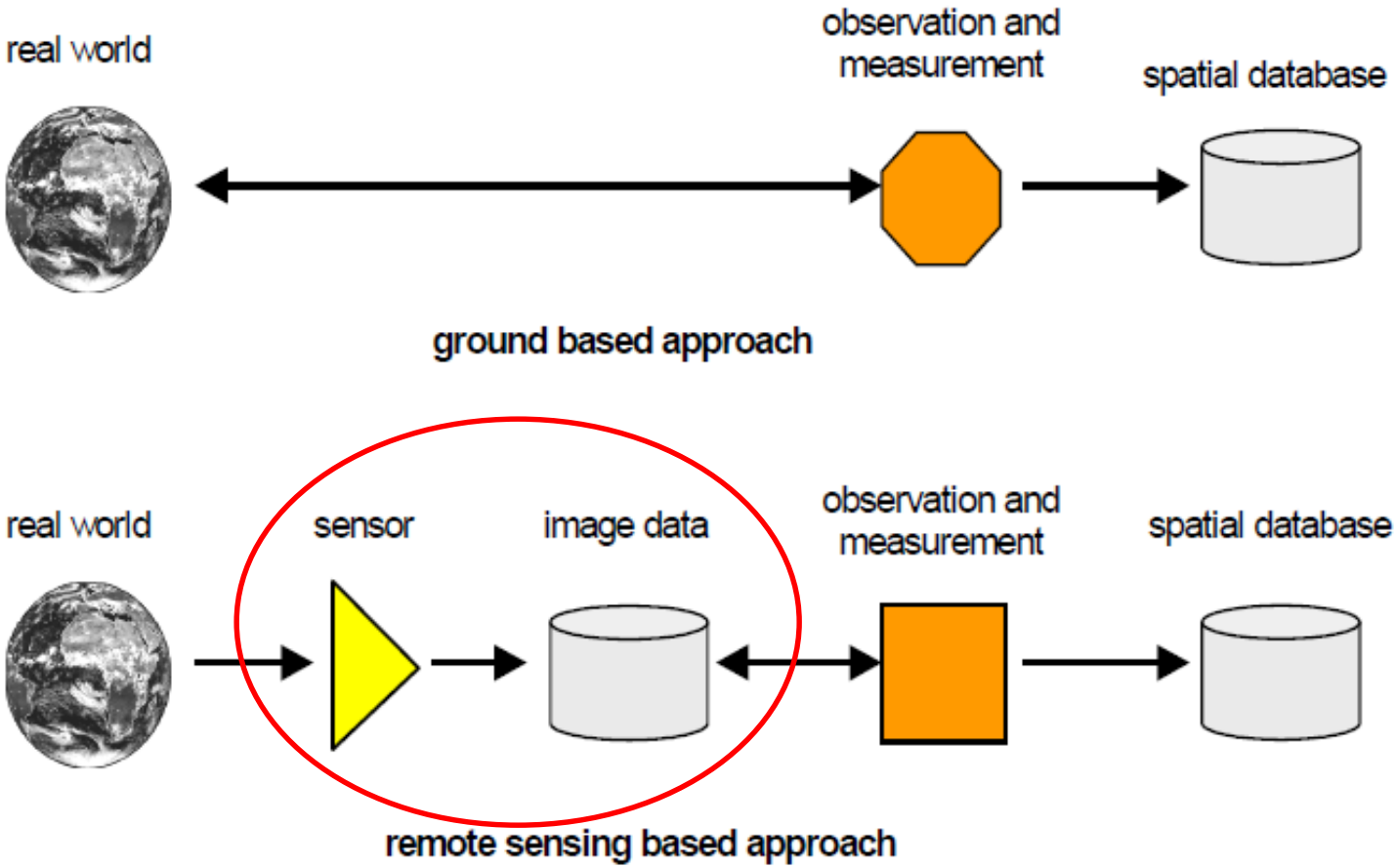
- **Data**: Representation that can be operated upon
- **Information:**
- Data that has been interpreted by human beings
- Data once interpreted become information

Data and Geospatial Data ?

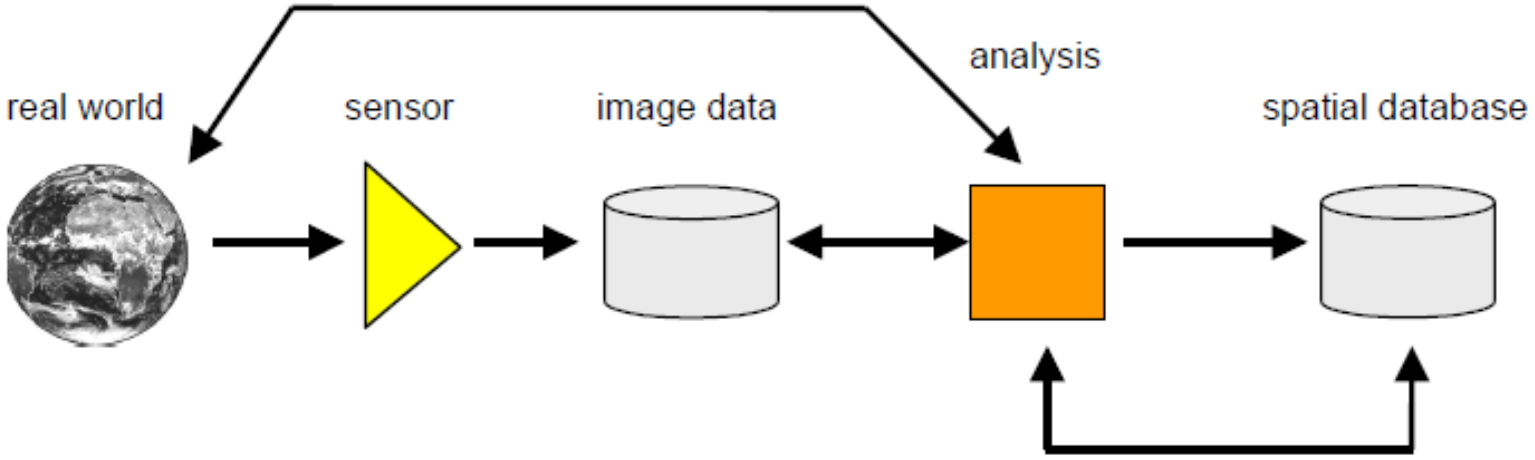
Geospatial Data Sources

- Field/ground Surveys
- Photography- Ground and Aerial
- Remote Sensing (Imaging: Space and airborne)
- Positioning Systems

Two Main Methods of Rs Data Collection



Integrated Data Acquisition



Combination of Ground-based and RS Techniques

Principle

- A remote sensing satellite is an artificial satellite designed to observe Earth from orbit and collect information about its surface, atmosphere, and oceans.
- Equipped with various sensors and devices, these satellites capture images from space in different parts of the electromagnetic spectrum, including visible light, infrared, and microwave wavelengths.

Principle

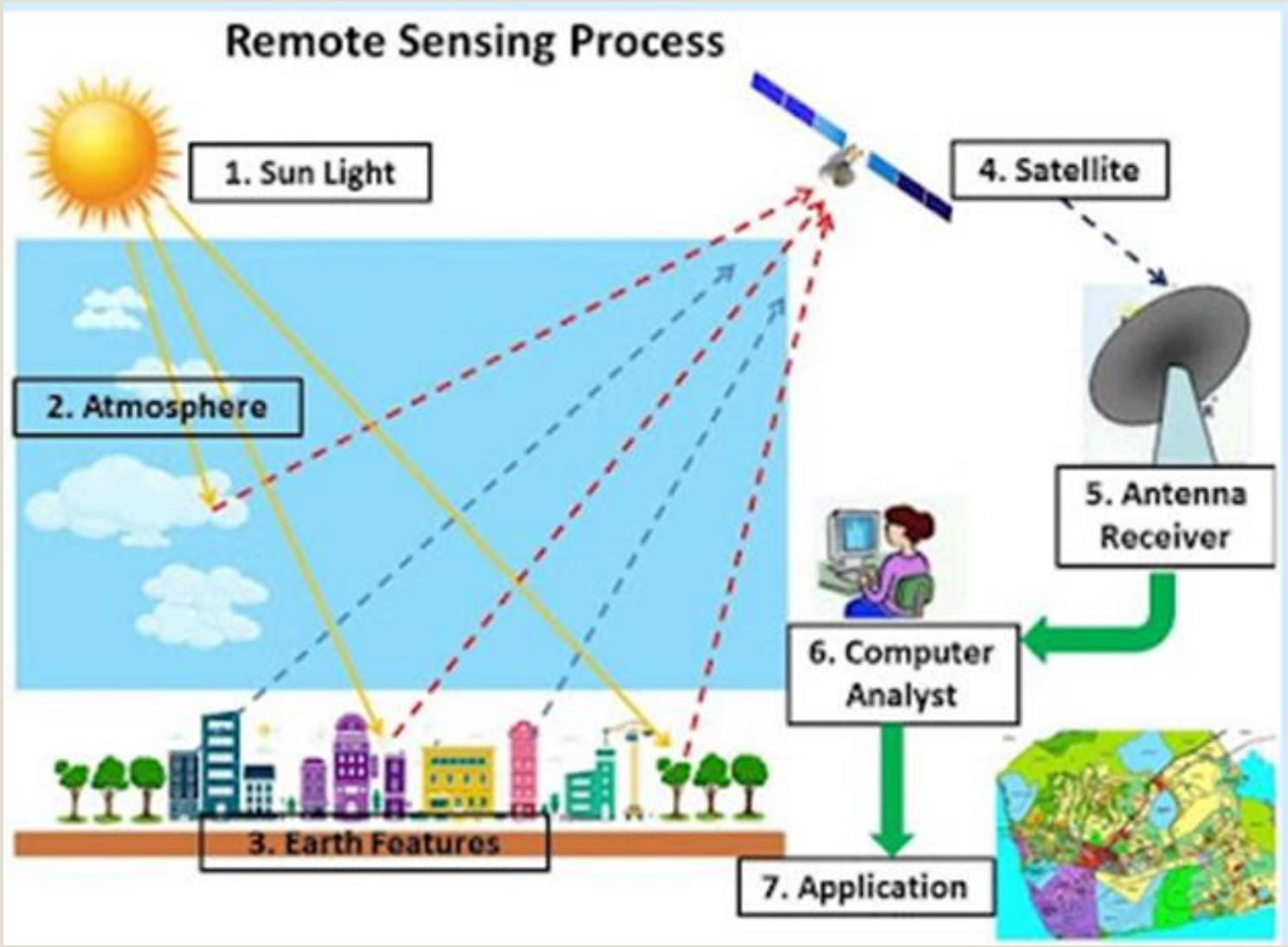
- RS largely concerns the measurement of EM energy which is reflected, scattered or emitted by different earth surface objects
- Different objects return different kinds and amount of energy, detecting these differences enables their identification

Process

- Energy Source or Illumination
- Radiation and the Atmosphere
- Radiation's Interaction with the Target
- Recording of Energy by the Sensor
- Transmission, Reception, and Processing
- Interpretation and Analysis
- Application

Earth Observation System

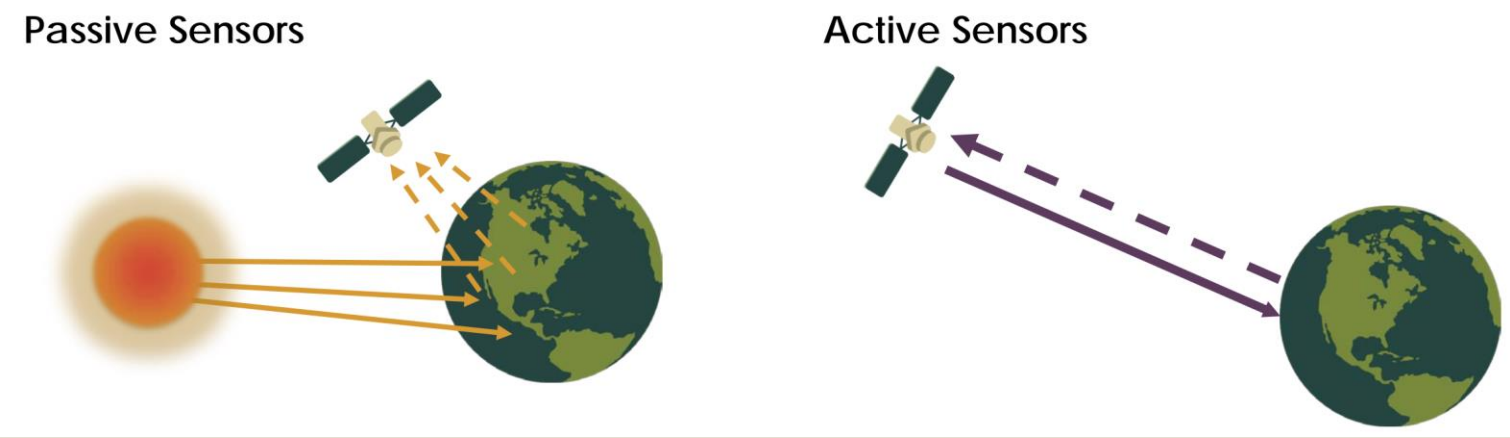
- Scene
  - Earth Surface and the Atmosphere
- RS Systems
  - Sensors
  - Platforms
- Processing/Analysis


Remote Sensing Process
1. Sun Light
2. Atmosphere
3. Earth Features
4. Satellite
5. Antenna Receiver
6. Computer Analyst
7. Application

Remote Sensors

- The detection and recording instruments for RS are known as ***remote sensors***
- These are carried on aircrafts and earth orbiting spacecrafts, which has led to the familiar phrase

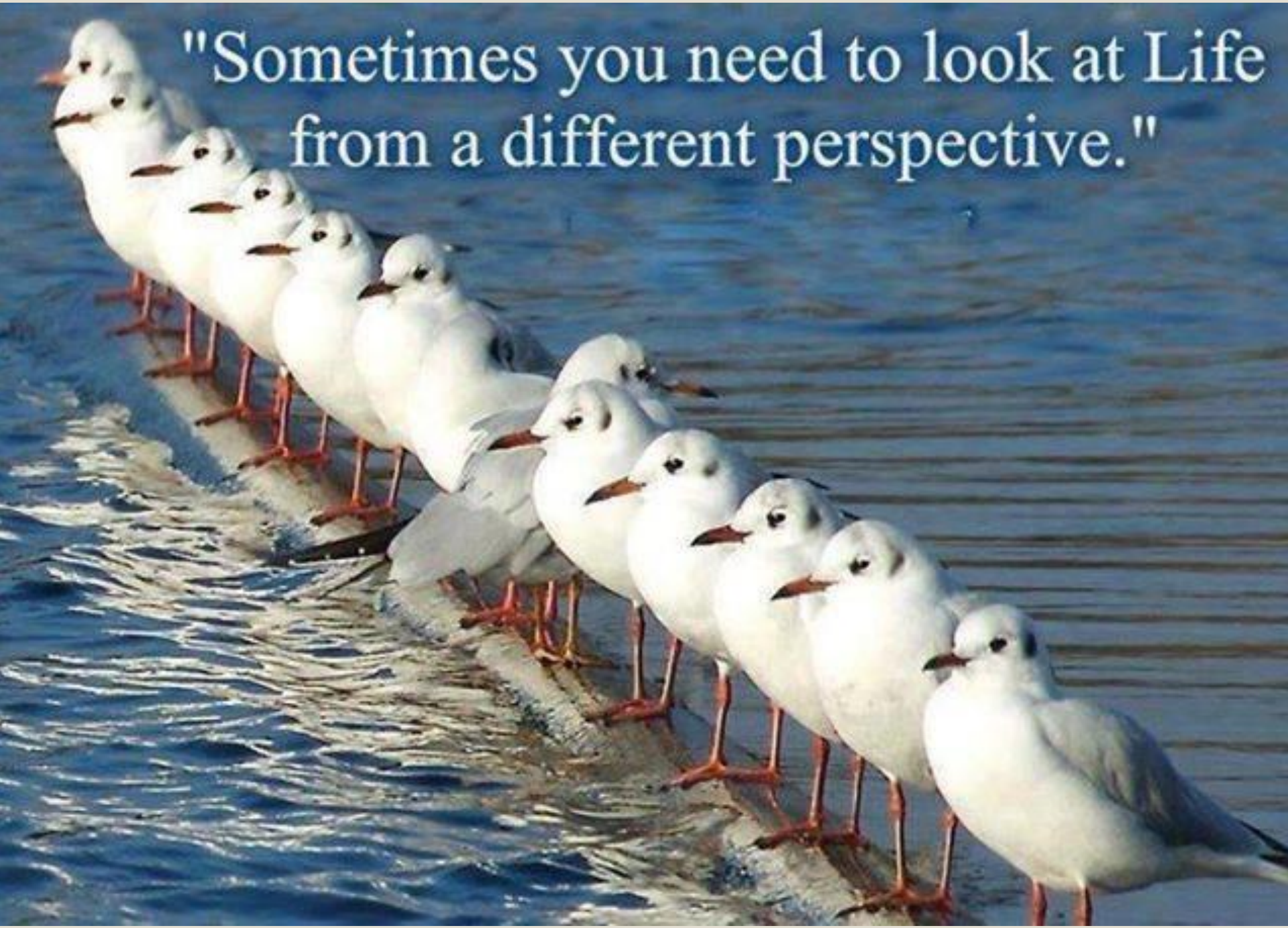
"Eye in the sky"



Data Analysis

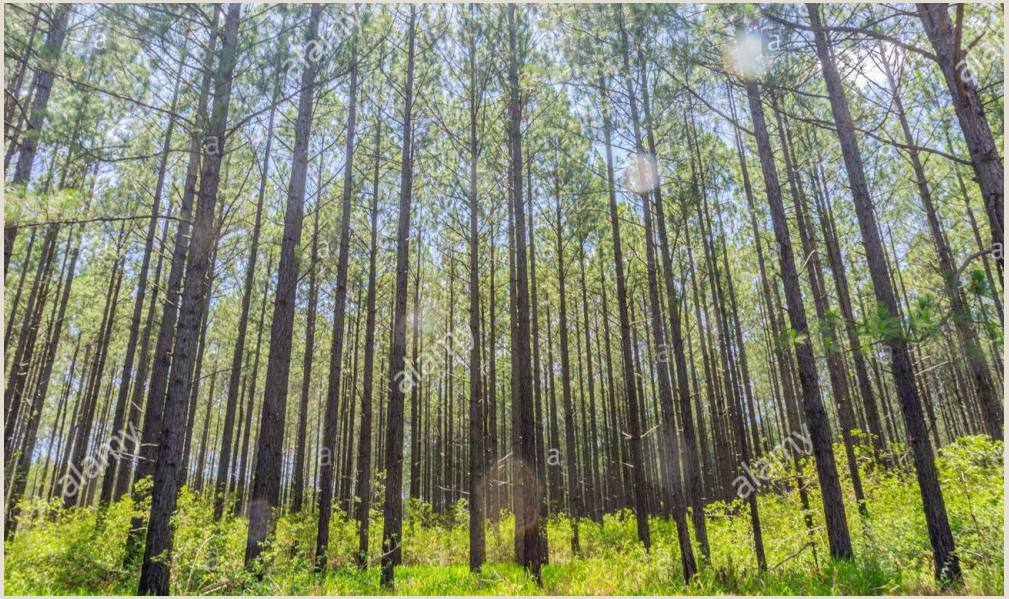
- Specialized knowledge is required to analyze RS data due to
  - Image presentation
  - Unusual scales and resolutions
  - Unfamiliar Overhead view
  - Use of several regions of EM spectrum (out side visible)

Look Perspective?


"Sometimes you need to look at Life
from a different perspective."

Overhead view (looking perspective)








ISIS

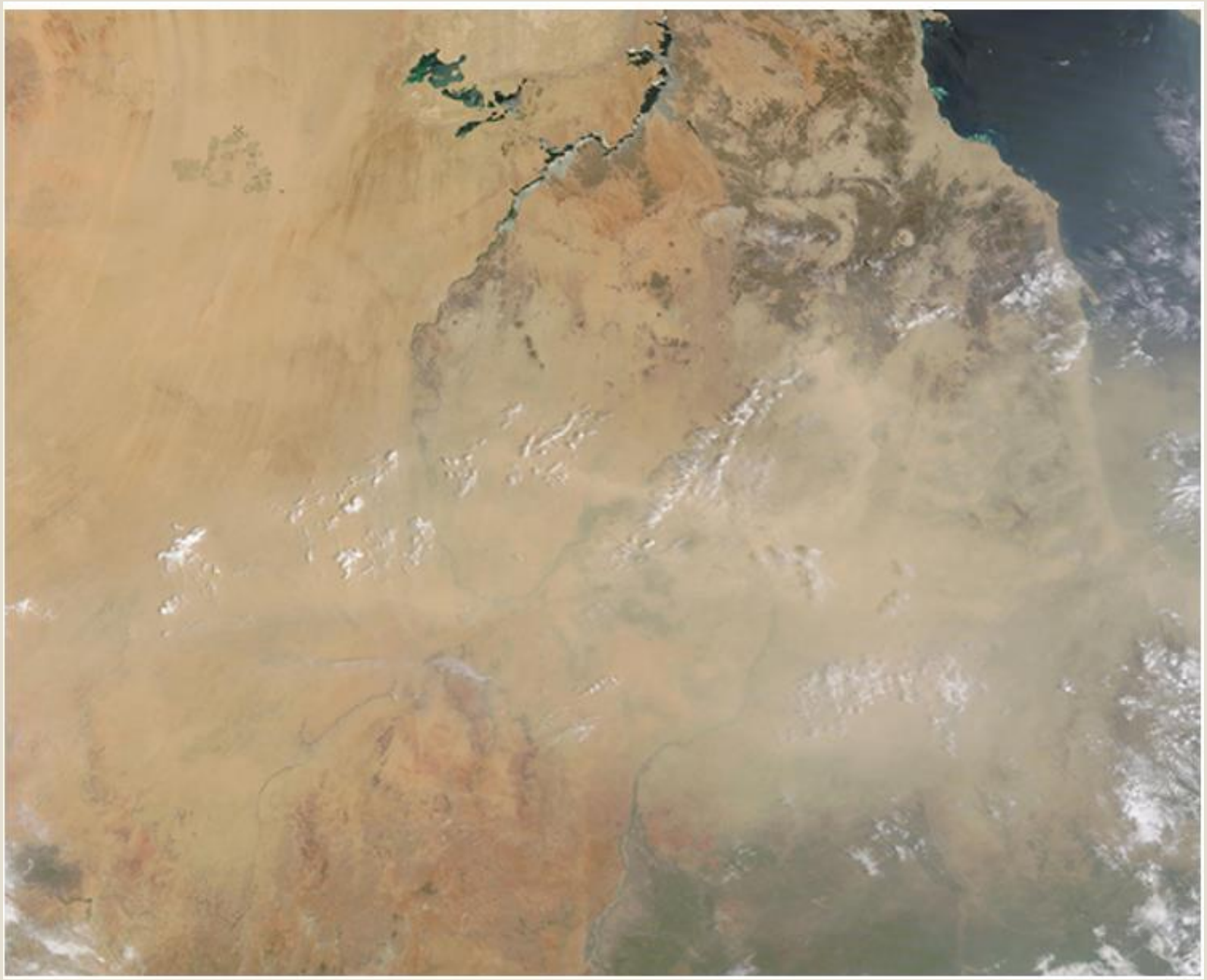



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Advantages of RS

- Large area coverage enabling global, regional surveys of variety of themes
- Repetitive coverage, allowing monitoring of dynamic themes like water, agriculture, etc
- Data acquisition over inaccessible areas
- Data acquisition over multiple heights allowing different scales and resolutions
- Fast and accurate data processing

Large Area Coverage



Repeated Coverage

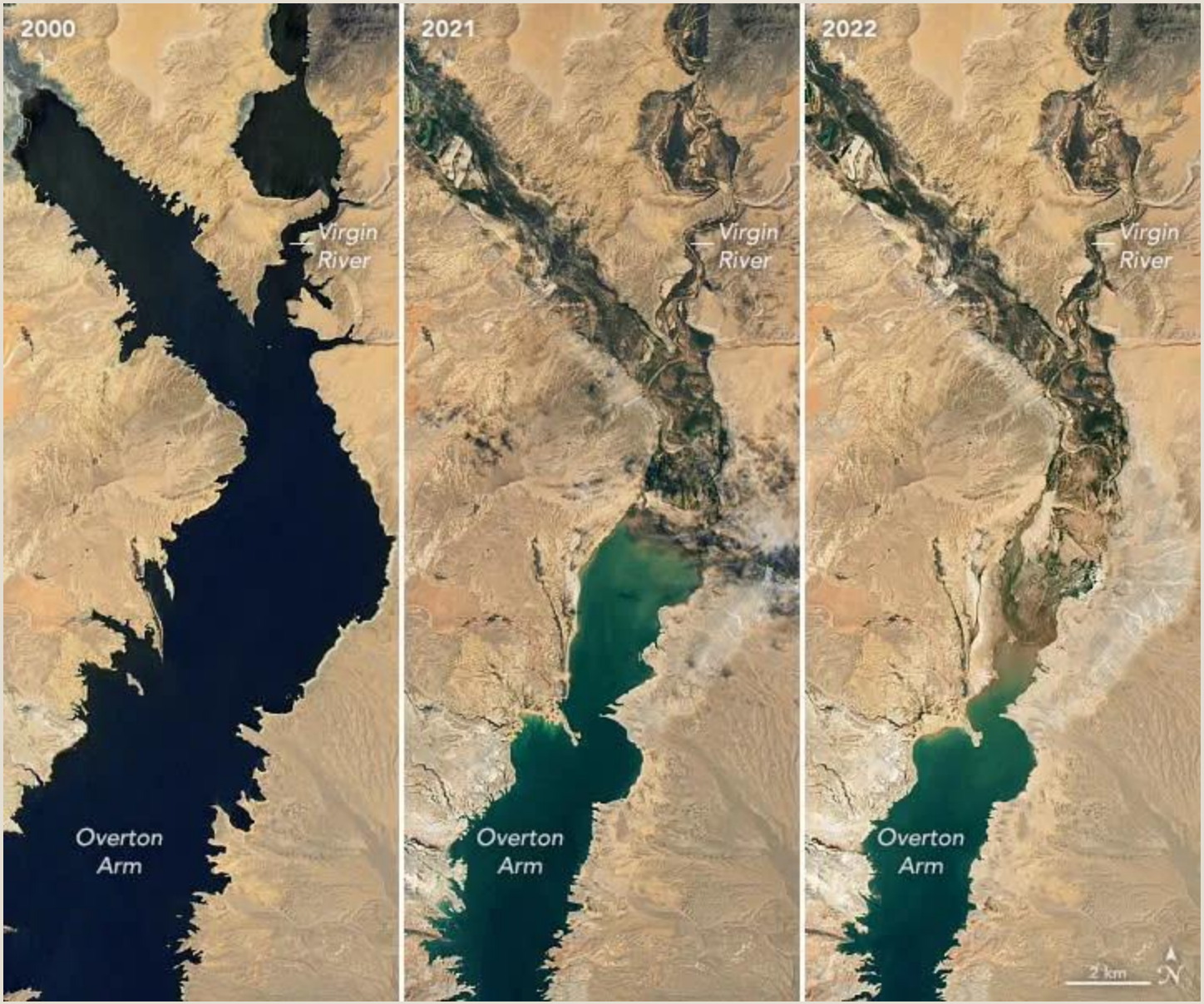


July 2006



June 2017

50


2000
Virgin River
Overton Arm
2021
Virgin River
Overton Arm
2022
Virgin River
Overton Arm
2 km
N

Inaccessible Area Coverage



Antarctica- Penguin's Colonies

Limitations of RS

- Often oversold, it cannot provide all the information needed
- Human interaction: Human beings select the appropriate remote sensing system to collect the data, set specifications and how the data are processed, this may introduce errors
- Remote sensing instruments may become uncalibrated, resulting in errors in the data
- Powerful active remote sensor systems that emit their own electromagnetic radiation can affect the phenomenon being investigated

Limitations of RS

- Remote sensor data may be expensive to collect , Hopefully, the information extracted from the remote sensor data justifies the expense
- The people who analyze the data are often the most expensive part of the remote sensing system

Earth Resource Analysis Perspective

- **RS data and digital image processing is used for many applications**

- **Our domain focuses on the art and science of applying remote sensing digital image processing for the extraction of useful Earth resource information**

Earth Resource Analysis Perspective

- **Earth Resource Information**
  - **Any information concerning terrestrial vegetation, soils, minerals, rocks, water, certain atmospheric characteristics, and urban infrastructure**

Application OF RS

- Successful application of any RS system is " the person or persons" using this data
- RS Data is just raw but when processed by a human interpreter, becomes "the usable info" only, and its applications depends upon how a person :-
  - Understands it
  - Knows how to interpret
  - Know how best to use it

APPLICATION OF RS

- A thorough understanding of the problem is essential to the productive application of any RS data
- Also, no single combination of the data acquisition and analysis procedures satisfy the needs of all data users

Applications of RS

- For a successful application we must have:-
  - A clear definition of the problem at hand
  - Evaluation of the potential for addressing the problem with RS techniques
  - Identification of the RS data acquisition procedures appropriate to the task
  - Determination of the data interpretation technique
  - Requirement and need of other reference data
  - Identification of the criteria by which the quality of collected information can be judged

Applications

- Regional planning and site investigations
  - Topographic and geologic considerations
  - Crossing sites, bridges, pipelines(slopes)
  - Site selections
- Water resource- availability, (irrigation, drinking, recreation, pollution, flooding)
- Highway alignment
- Terrain Mapping

Water Resource Applications

- To monitor :-
  - Quality
  - Quantity
  - Geographic distribution
- Flood damage estimates
- Flood extent and progress, relief
- Water pollution detection
  - Domestic waste
  - Industrial waste
  - Locate the discharge and extent of its plume
  - Sediment pollution
  - Oil spills

Urban and Regional Planning

- Population estimate
- Housing quality assessment
- Traffic and parking studies
- Specific site selection
- Urban change detection (urban development, sprawl)

Applications

- Earth observation,
- Environmental monitoring,
- Mapping and navigation,
- Meteorology,
- Cartography,
- Mineral resource exploration,
- Insurance,
- Defence, and security,
- Flood and drought monitoring,
- Soil moisture assessment,
- Vegetation analysis,
- Deforestation tracking,
- Forest fire detection,

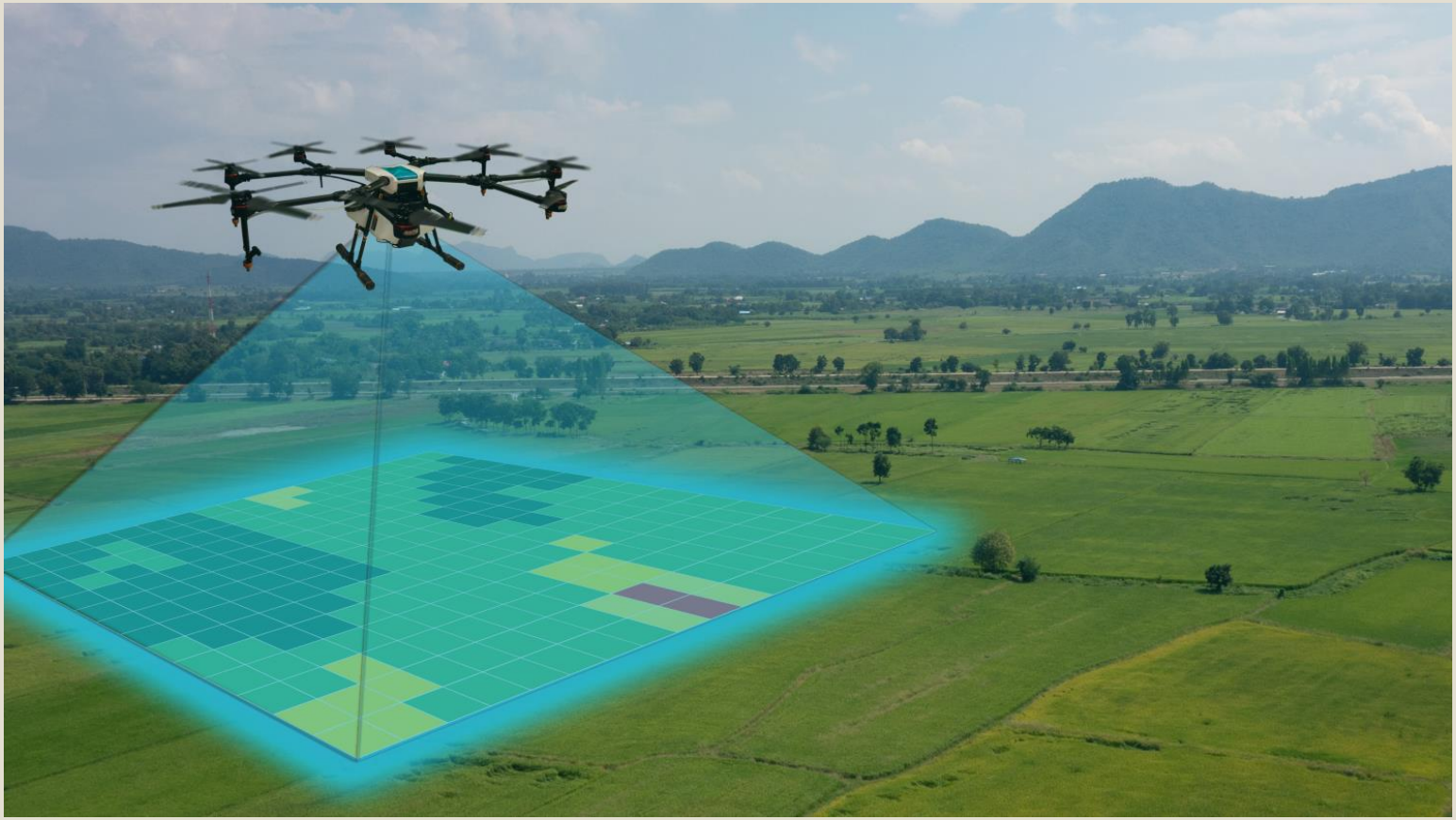
Updating Urban Maps



Natural Disaster Management

- Wildfires
- Floods
- Severe storms
- Volcanic eruption
- Earthquakes
- Hurricanes

Agricultural Monitoring



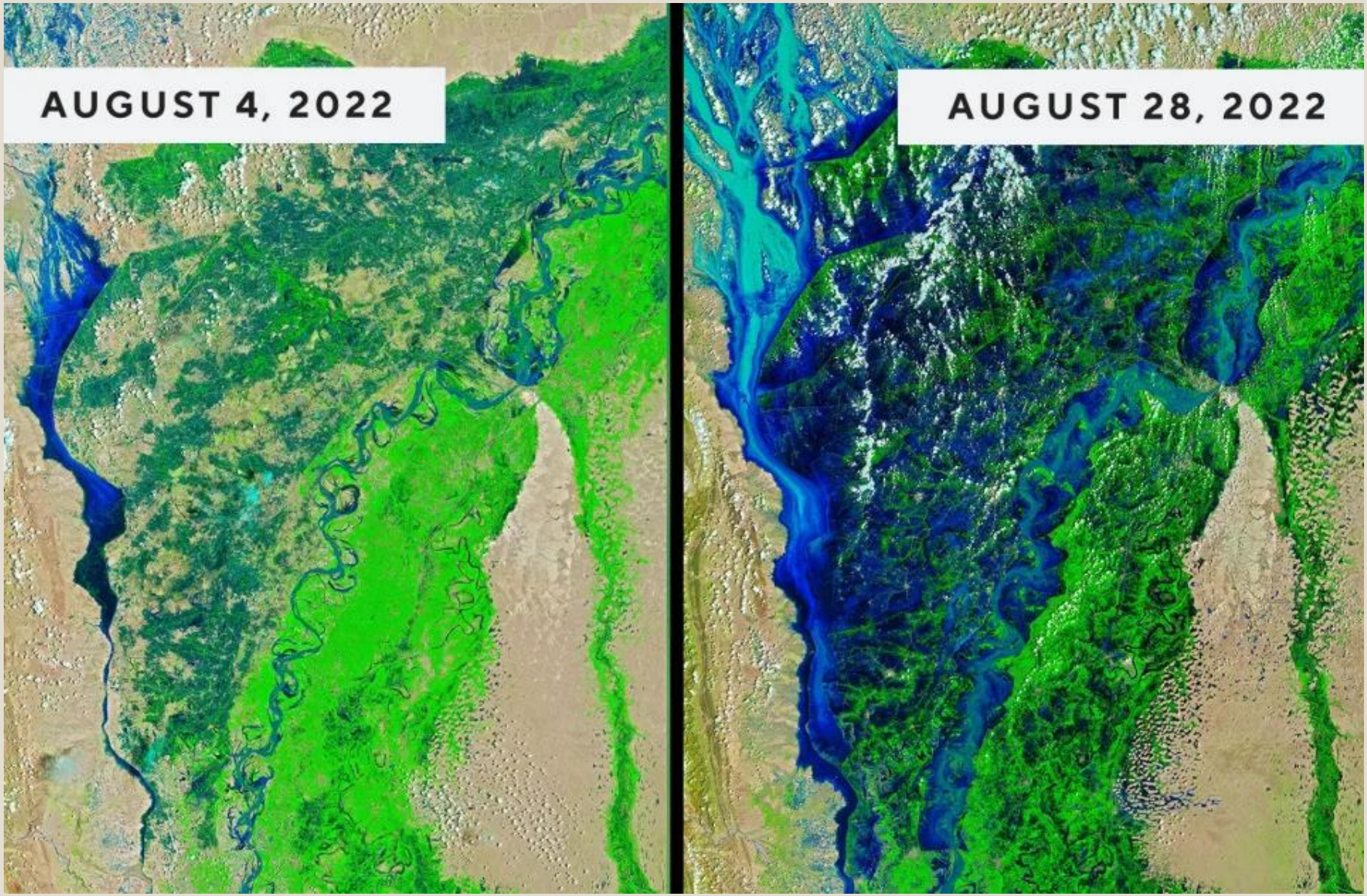
Smart City Project

Remote sensing satellites assist authorities in zoning, urban planning, infrastructure modelling, security, and law enforcement planning.



Forest Fires -Brazil




AUGUST 4, 2022
AUGUST 28, 2022

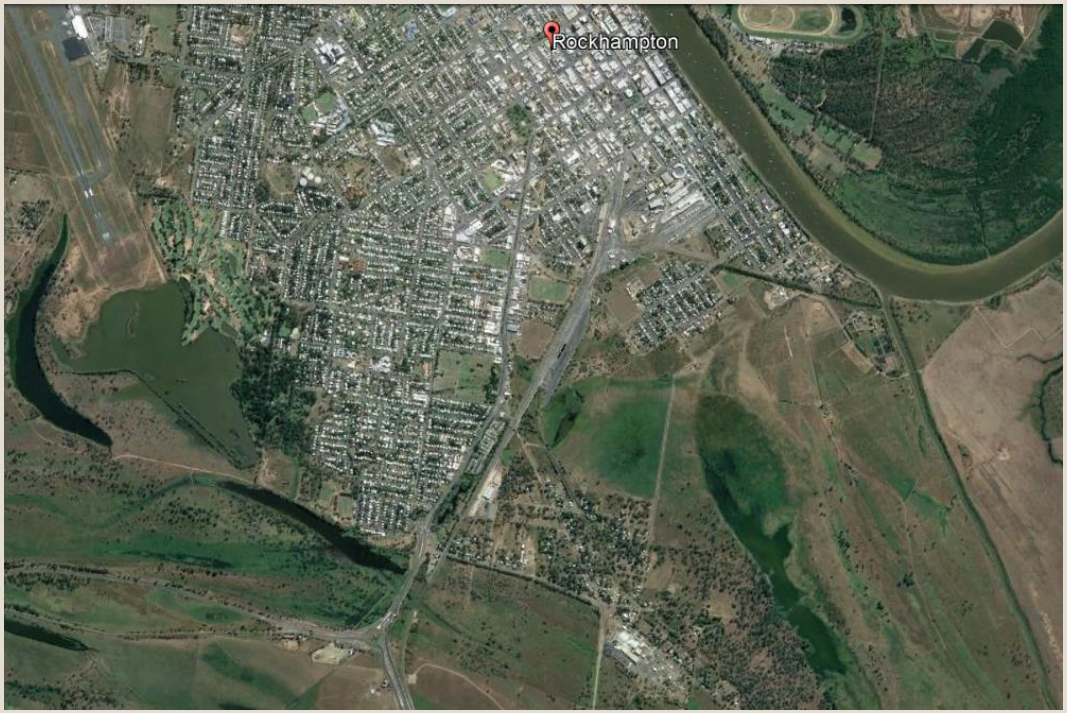
Derna Libya



Pre Flood Sep. 2, 2023

Post Flood Flood Sep. 12, 2023


airport
Rockhampton
—runways—
Fitzroy River
Bruce Highway—
1 km
N


Rockhampton

Severe Storms-Tornados



Volcanic Eruptions



Earthquake Damage Assessment



Roof collapse



Partial collapse

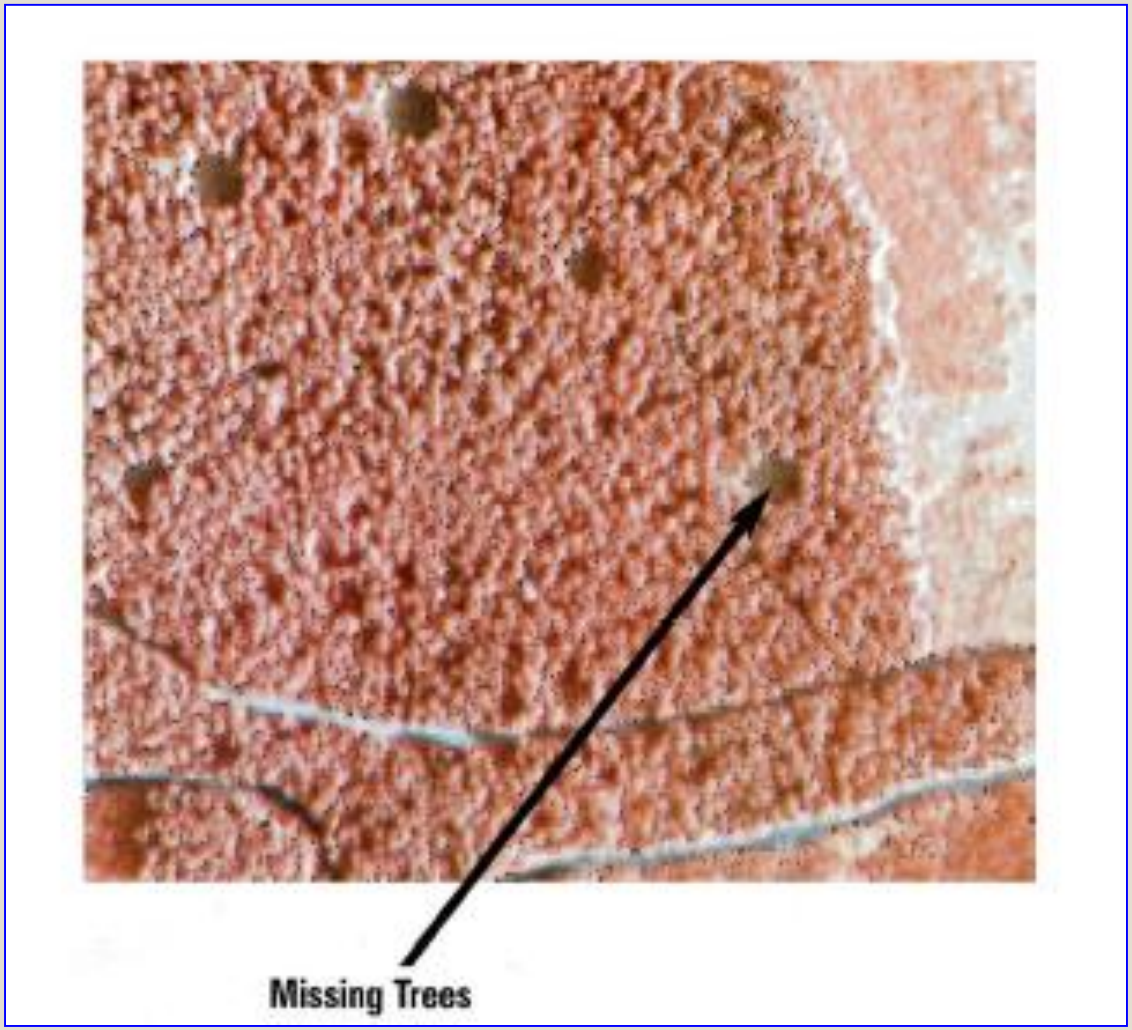


Complete damage

Copernicus Sentinel-3 image - Hurricane Dorian as it pummels the Bahamas on Sept. 2, 2019



Monitoring Of Illegal Logging



Hurricane Lee Atlantic Ocean-Catastrophic damage will occur

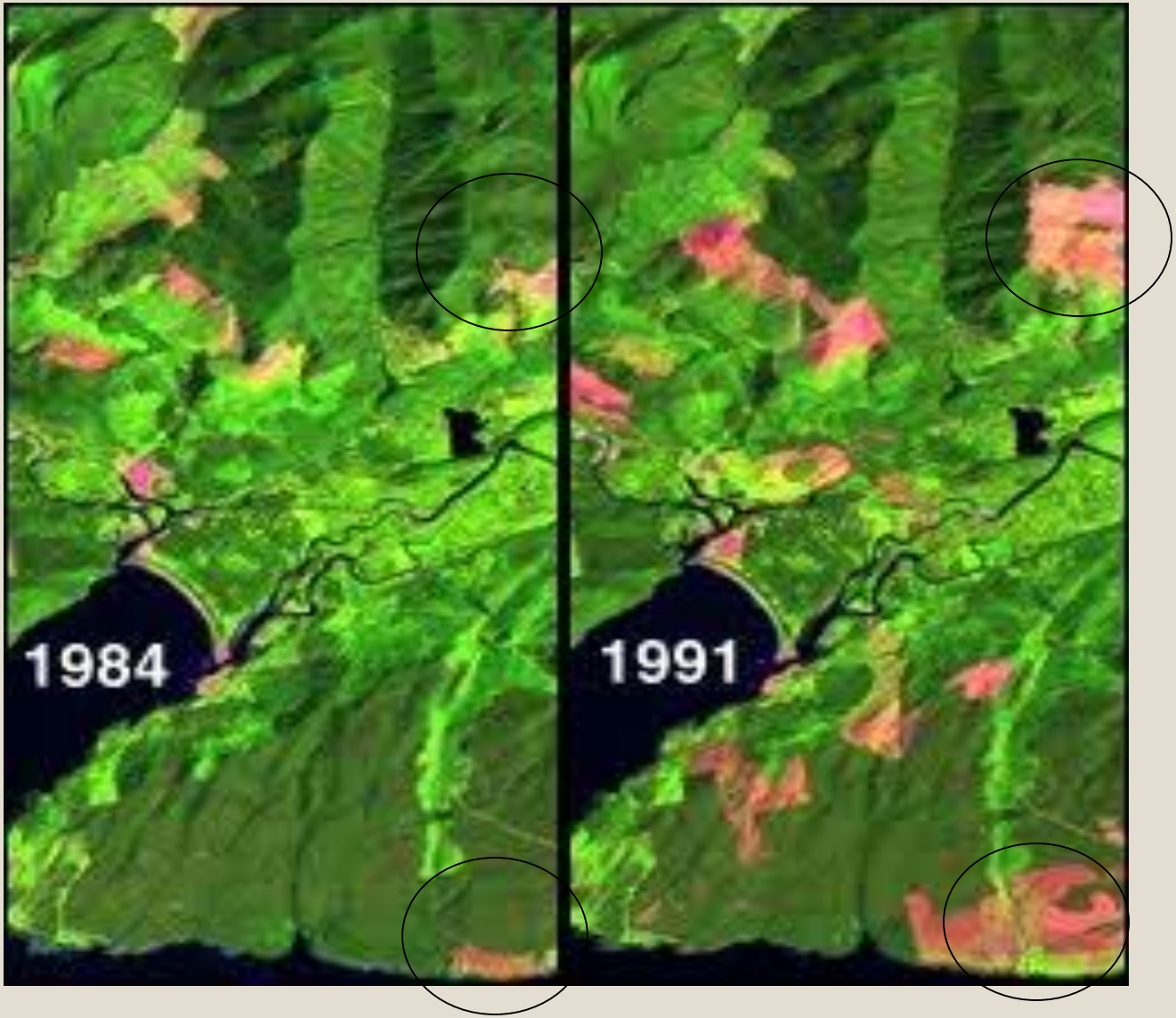


Sep 12, 2023



Sep 6, 2023

Forestry - Change Detection



Desertification and Drought




HARAD GAS PLANT, SAUDI ARABIA · September 14, 2019


Damage to Oil/Gas Infrastructure at Abqaiq – Image 2
15 Sep 19
N
Impact point
© 2019 DigitalGlobe



Imam Khomeini Space Center in Iran



Imam Khomeini Space Center in Iran

Moon Bay Ct

Remains of missing man found on Google Earth after 20 years after submerged car spotted in lake



Applications OF RS

- It is so broad in its applications that nobody owns the field and benefits are derived by all for basic research or operational applications
- RS is and will continue to play an increasingly broad and imp role in inventorying, monitoring and managing earth resources

Popular Uses/Users of RS (USA)

- Climate and weather forecasting (NOAA)
- Tracking Storms, National Weather Systems (NWS)
- Exploring Earth and other planets (NASA)
- Aviation and flight control (FAA)
- Precision farming and Economics (USDA)
- Military Intelligence (DoD)

Reading Assignment

- **Chapter 1, Introduction to RS, by James B. Campbell**

